Lattice Crypto Simulator

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Introduction

The goal of this project is to create a script for encrypting and decrypting an array of binary through quantum-safe bit-wise Learning With Errors (**LWE**) lattice cryptography techniques. The overarching aim of this project is to learn more about the inner workings of lattice based encryption and better understand its implementation in the future of computing.

Resources used

- Python as the language
- Numpy for arithmetic computations
- Matlablib for displaying useful information out to a graph
- Kyber-512 for real-world variable values and its small message optimization

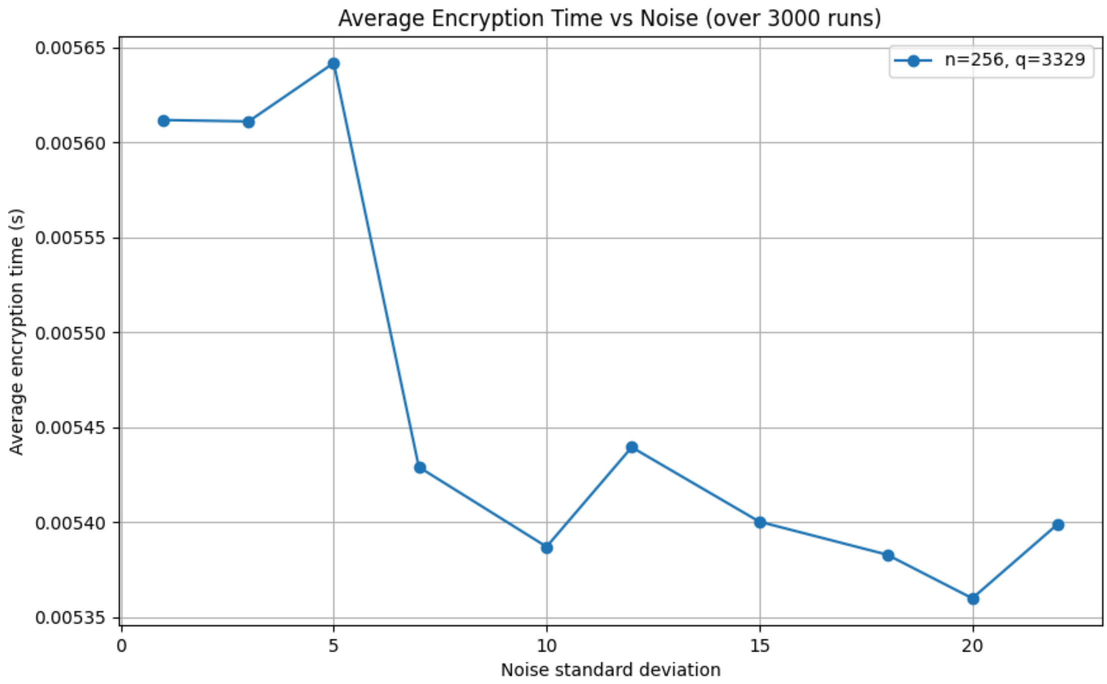
The Basics

In this section I would like to go over the basic principles that govern lattice's, why they work for this application, and how it gets used in LWE.

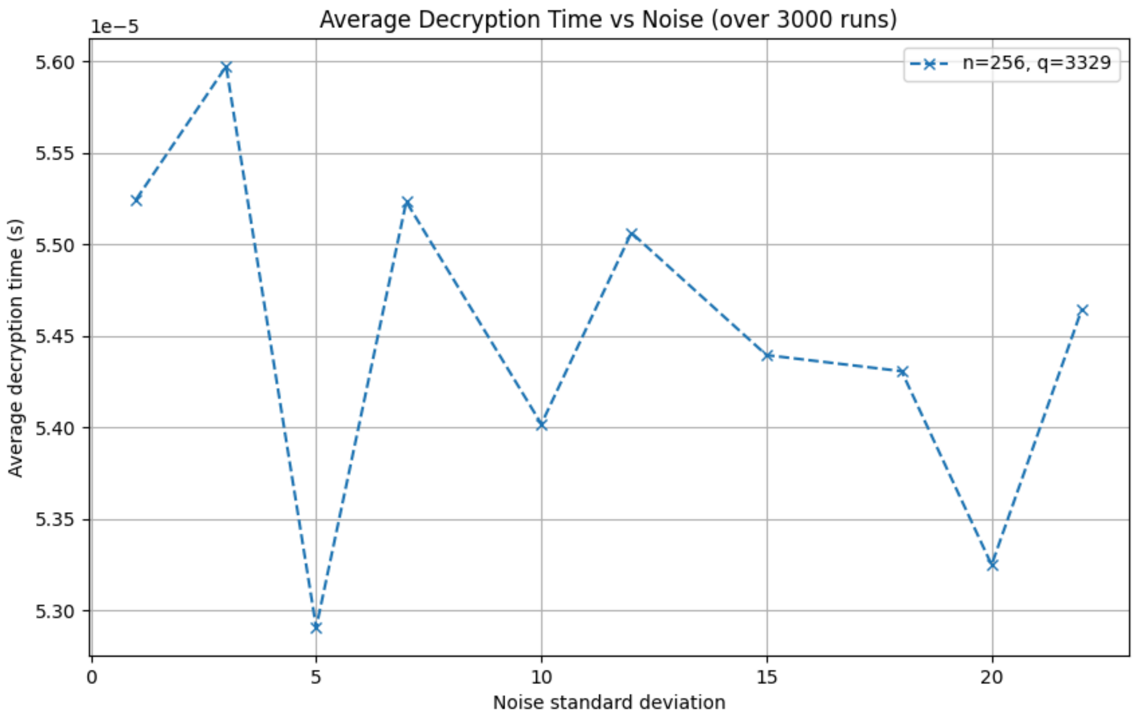
Keywords:

- Lattice - A repeating series of points formed by using integer amounts of *n* number of linearly unique vectors
- Basis Vectors - Linearly unique vectors that generate a lattice
- Shortest Vector Problem (SVP) - The problem of finding the shortest basis vector that makes up a lattice
- Closest Vector Problem (CVP) - The problem of trying to find the closest lattice point to any given point in space

Testing Graphs



256 dimensions and 3329 q value encoding time over noise standard deviation amounts



256 dimensions and 3329 q value decoding time over noise standard deviation amounts

Challenges

During the research, presentation, and demo development for this project; there were many challenges that were faced. These include (*but are certainly not limited to*):

- Learning the underlying mathematical principles that make it possible
- CPU Cache bloating which resulted in unexpected RNG number generation by directly affecting the entropy source
- RNG normalization which resulted in the random key generation failing
- Memory limitations due to a lack of optimization which limited testing
- Lack of python knowledge which limited the progress made during testing

Lessons

The main goal of this presentation is to gain a better understanding of a public key cryptography technique by presenting, and creating a working demo of, an idea. To that end, the main lessons taken from this were:

- Showed how lattice encryption works at a basic level
- Showed future steps of cryptography against a quantum computing age
- Showed limitations of lattice cryptography in the future

Future plans

This is only a very basic outline of what a real LWE technique would look like and falls short of existing security and modern lattice encryption techniques. In the future I plan to use the material outlined in this presentation to:

- Implement a polynomial based method of representing the vectors
- Including padding for added security
- Include message signatures to increase sender/reciver confidence
- Allow for communications over serial lines or radio to add a realistic use-case to the software

References

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